

Understanding Architectural Approaches for MLOps Systems: SMS External Material

Pedro Delgado
University of São Paulo
São Carlos, Brazil
pedro.delgado@usp.br

Giovanna Belasco
University of São Paulo
São Carlos, Brazil
giovannabelasco@usp.br

Damian Tamburri
University of Sannio
Benevento, Italy
d.a.tamburri@tue.nl

Elisa Yumi Nakagawa
University of São Paulo
São Carlos, Brazil
elisa@icmc.usp.br

This document presents the external material associated with the systematic mapping study (SMS) on the identification of architectural approaches applied to MLOps systems. It primarily provides the information needed to support the reproducibility and auditability of the SMS, including a set of files containing the raw data.

SMS Goal: This SMS's goal is to analyze primary studies to identify architectural approaches applied to MLOps systems in the scientific literature.

Research question (RQ): This SMS has one RQ—"What architectural approaches have been proposed and applied in MLOps systems?". The rationale for this question is that a better understanding of the diverse architectural approaches to MLOps systems could support practitioners in making more informed decisions and researchers with potential topics for future investigations.

Search string: (*mlops* OR "*ml ops*" OR "*machine learning operations*" OR "*ml operations*" OR "*machine learning engineering*" OR "*ml engineering*" OR *modelops* OR "*model ops*" OR "*model operations*") AND ("*design pattern*" OR "*architectural pattern*" OR "*architecture pattern*" OR "*architectural tactic*" OR "*architecture tactic*" OR "*architectural decision*" OR "*architecture decision*" OR "*architectural design decision*" OR "*software architecture*" OR *quality*)

Selection Criteria: The selection criteria are composed of one inclusion criteria (IC) and six exclusion criteria (EC):

- **IC1:** Study addresses an architectural approach for MLOps systems.
- **EC1:** Study does not address any architectural approach in the context of MLOps systems.
- **EC2:** Full text is not available.
- **EC3:** Study is written in a language other than English.
- **EC4:** Study is not peer-reviewed (e.g., tutorial, poster, or keynote abstract).
- **EC5:** Study is a duplicate or a shorter version of a more complete study already included.
- **EC6:** Study is a secondary study (e.g., literature review).

Publication databases: The databases used in this SMS are:

- Scopus (<https://www.scopus.com>)

- ACM Digital Library (<https://dl.acm.org>)
- IEEE Xplore (<https://ieeexplore.ieee.org/Xplore>)
- ScienceDirect (<https://www.sciencedirect.com>)
- Wiley (<https://onlinelibrary.wiley.com>)

Search String Configuration: We configured the search string for the search engine of each publication database and applied year filters, as described in the paper.

- **Scopus:** *TITLE-ABS-KEY* ("*mlops*" OR "*ml ops*" OR "*machine learning operations*" OR "*ml operations*" OR "*machine learning engineering*" OR "*ml engineering*" OR "*modelops*" OR "*model ops*" OR "*model operations*") AND ("*design pattern*" OR "*architectural pattern*" OR "*architecture pattern*" OR "*architectural tactic*" OR "*architecture tactic*" OR "*architectural decision*" OR "*architecture decision*" OR "*architectural design decision*" OR "*software architecture*" OR "*quality*") AND *PUBYEAR* > 2014 AND *PUBYEAR* < 2027
 - **Filter:** 2015 to 2026
 - **Search date:** January 15, 2026
 - **Studies retrieved:** 278
- **IEEE Xplore:** (((("Document Title": "*mlops*" OR "Document Title": "*ml ops*" OR "Document Title": "*machine learning operations*" OR "Document Title": "*ml operations*" OR "Document Title": "*machine learning engineering*" OR "Document Title": "*ml engineering*" OR "Document Title": "*modelops*" OR "Document Title": "*model ops*" OR "Document Title": "*model operations*") OR ("Abstract": "*mlops*" OR "Abstract": "*ml ops*" OR "Abstract": "*machine learning operations*" OR "Abstract": "*ml operations*" OR "Abstract": "*machine learning engineering*" OR "Abstract": "*ml engineering*" OR "Abstract": "*modelops*" OR "Abstract": "*model ops*" OR "Abstract": "*model operations*") OR ("Author Keywords": "*mlops*" OR "Author Keywords": "*ml ops*" OR "Author Keywords": "*machine learning operations*" OR "Author Keywords": "*ml operations*" OR "Author Keywords": "*machine learning engineering*" OR "Author Keywords": "*ml engineering*" OR "Author Keywords": "*modelops*" OR "Author

Keywords": "model ops" OR "Author Keywords": "model operations")) AND (("Document Title": "design pattern" OR "Document Title": "architectural pattern" OR "Document Title": "architecture pattern" OR "Document Title": "architectural tactic" OR "Document Title": "architecture tactic" OR "Document Title": "architectural decision" OR "Document Title": "architecture decision" OR "Document Title": "architectural design decision" OR "Document Title": "software architecture" OR "Document Title": "quality") OR ("Abstract": "design pattern" OR "Abstract": "architectural pattern" OR "Abstract": "architecture pattern" OR "Abstract": "architectural tactic" OR "Abstract": "architecture tactic" OR "Abstract": "architectural decision" OR "Abstract": "architecture decision" OR "Abstract": "architectural design decision" OR "Abstract": "software architecture" OR "Abstract": "quality") OR ("Author Keywords": "design pattern" OR "Author Keywords": "architectural pattern" OR "Author Keywords": "architecture pattern" OR "Author Keywords": "architectural tactic" OR "Author Keywords": "architecture tactic" OR "Author Keywords": "architectural decision" OR "Author Keywords": "architecture decision" OR "Author Keywords": "architectural design decision" OR "Author Keywords": "software architecture" OR "Author Keywords": "quality"))))

- **Filter:** 2015 to 2026
- **Search date:** January 15, 2026
- **Studies retrieved:** 68

- **ACM Digital Library:** [[Title: "mlops"] OR [Title: "ml ops"] OR [Title: "machine learning operations"] OR [Title: "ml operations"] OR [Title: "machine learning engineering"] OR [Title: "ml engineering"] OR [Title: "modelops"] OR [Title: "model ops"] OR [Title: "model operations"] OR [Abstract: "mlops"] OR [Abstract: "ml ops"] OR [Abstract: "machine learning operations"] OR [Abstract: "ml operations"] OR [Abstract: "machine learning engineering"] OR [Abstract: "ml engineering"] OR [Abstract: "modelops"] OR [Abstract: "model ops"] OR [Abstract: "model operations"] OR [Keywords: "mlops"] OR [Keywords: "ml ops"] OR [Keywords: "machine learning operations"] OR [Keywords: "ml operations"] OR [Keywords: "machine learning engineering"] OR [Keywords: "ml engineering"] OR [Keywords: "modelops"] OR [Keywords: "model ops"] OR [Keywords: "model operations"]] AND [[Title: "design pattern"] OR [Title: "architectural pattern"] OR [Title: "architecture pattern"] OR [Title: "architectural tactic"] OR [Title: "architecture tactic"] OR [Title: "architectural decision"] OR [Title: "architecture decision"] OR [Title: "architectural design decision"] OR [Title: "software architecture"] OR [Title: "quality"] OR [Title: "quality"] OR [Abstract: "design pattern"] OR [Abstract: "architectural pattern"] OR [Abstract: "architecture pattern"] OR [Abstract: "architectural tactic"] OR [Abstract: "architecture tactic"] OR [Abstract: "architectural decision"] OR [Abstract: "architecture decision"] OR [Abstract: "architectural design decision"] OR [Abstract: "software architecture"] OR [Abstract: "quality"] OR [Keywords: "design pattern"] OR [Keywords: "architectural pattern"] OR [Keywords: "architecture pattern"] OR [Keywords: "architectural tactic"] OR [Keywords: "architecture tactic"] OR [Keywords: "architectural decision"] OR [Keywords: "architecture decision"] OR [Keywords: "architectural design decision"] OR [Keywords: "software architecture"] OR [Keywords: "quality"]]] AND [E-Publication Date: (01/01/2015 TO 12/31/2026)]

ware architecture"] OR [Title: "quality"] OR [Abstract: "design pattern"] OR [Abstract: "architectural pattern"] OR [Abstract: "architecture pattern"] OR [Abstract: "architectural tactic"] OR [Abstract: "architecture tactic"] OR [Abstract: "architectural decision"] OR [Abstract: "architecture decision"] OR [Abstract: "architectural design decision"] OR [Abstract: "software architecture"] OR [Abstract: "quality"] OR [Keywords: "design pattern"] OR [Keywords: "architectural pattern"] OR [Keywords: "architecture pattern"] OR [Keywords: "architectural tactic"] OR [Keywords: "architecture tactic"] OR [Keywords: "architectural decision"] OR [Keywords: "architecture decision"] OR [Keywords: "architectural design decision"] OR [Keywords: "software architecture"] OR [Keywords: "quality"]]] AND [E-Publication Date: (01/01/2015 TO 12/31/2026)]

- **Filter:** 2015 to 2026
- **Search date:** January 15, 2026
- **Studies retrieved:** 61

- **ScienceDirect:** Title, abstract, keywords: ("mlops" OR "machine learning operations" OR "ml engineering" OR "modelops") AND ("software architecture" OR "quality")

- **Filter:** 2015 to 2026
- **Search date:** January 15, 2026
- **Studies retrieved:** 21

- **Wiley Online Library:** [Title: (mlops OR "machine learning operations" OR "machine learning engineering") AND ("design" OR "architecture")] AND [Abstract: (mlops OR "machine learning operations" OR "machine learning engineering") AND ("design" OR "architecture")] AND [Keywords: (mlops OR "machine learning operations" OR "machine learning engineering") AND ("design" OR "architecture")]

- **Filter:** 2015 to 2026
- **Search date:** January 15, 2026
- **Studies retrieved:** 0

Files and replication package: All artifacts supporting this SMS are publicly available in a GitHub replication package: <https://github.com/DelgadoPedro/SAML-ICSA2026>. The repository includes (i) the complete lists of studies retrieved from each digital library, (ii) the study selection process, and (iii) the data extraction file containing raw data.